



The Oracle of Delphi: How the Ancient Greeks Relied on One Woman's Divine Visions

Privacy and Personal Information on Cloud Services

1.5.3. Team and Individual Contributions

- Each team must maintain a Git-based (e.g., GitHub, GitLab, ...) repository⁷ for the project. The repository must be associated with the individual accounts of all team members. The repository will be used to document the development of the project and to make individual contributions traceable. The repository should be used throughout the project rather than only immediately before a submission deadline. The lab instructor and/or marker may consider the repository's commit history, pull requests, issues, documentation, and other project records when assess individual contributions. The number of commits alone will not be used as a measure of contribution. The quality, significance, and nature of each member's contributions will also be considered.
- The graduate student serves as the project lead, responsible for coordinating the process, maintaining the overall project direction, and ensuring that the project develops toward a meaningful contribution. However, the project lead is also expected to be an active contributor, participating in activities such as problem formulation, literature review, methodology development, implementation, experimentation, analysis, and writing.
- Undergraduate members contribute to the project activities under the guidance of the project lead. All team members are expected to make substantive contributions, and no member should be limited solely to coordination or project management.
- As part of **Milestone 6. Project Management & Documentation (Graduate Student Only)**, the graduate student (project lead) is expected to document each member's contributions throughout the project, with a brief contribution statement provided at each major project milestone. These statements will be compiled into the final project management report.

Author

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Author

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1.5.4. Research Problem Selection

You may not select the same research problem as another group unless you are members of the same team. Once you have identified a potential research problem, communicate your choice to the lab instructor and obtain approval before proceeding to the next step.

⁶ <https://www.overleaf.com/read/bsrxtwqgggsb>

⁷ Web-based applications and cloud services may collect, use and/or disclose your personal information and store or access that information outside of Canada. Personal information includes, but not limited to, identifiable individual's information such as your name or your email address. At UVic, GitLab at gitlab.uvic.ca is highly recommended for coursework, research and development projects. For more information, see <https://www.uvic.ca/systems/services/pages/gitlab.php>. At your team's discretion, you may use GitHub to store information about your project, but we do not require nor recommend storing any personal identifying information. Also, we do not require public repo but it should be accessible to instructors and markers for evaluation purposes.



Team Project: Milestone 2 and GR List

Project

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1. Project
Definition and
Team Formation

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[2. Background and
Related Work](#)

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3. Solution Design
and
Implementation

4. Evaluation

5. Demo: 3-Minute
Talk (3MT)

[6. Project
Management and
Documentation
\(Grad Students
Only\)](#)

2. Background and Related Work



Assignment



Due October 8 at 5:00 AM



Please see the attached manual and sample submission. The sample is **for illustration only and is not a complete or authoritative example**. Please follow the manual for the actual requirements.

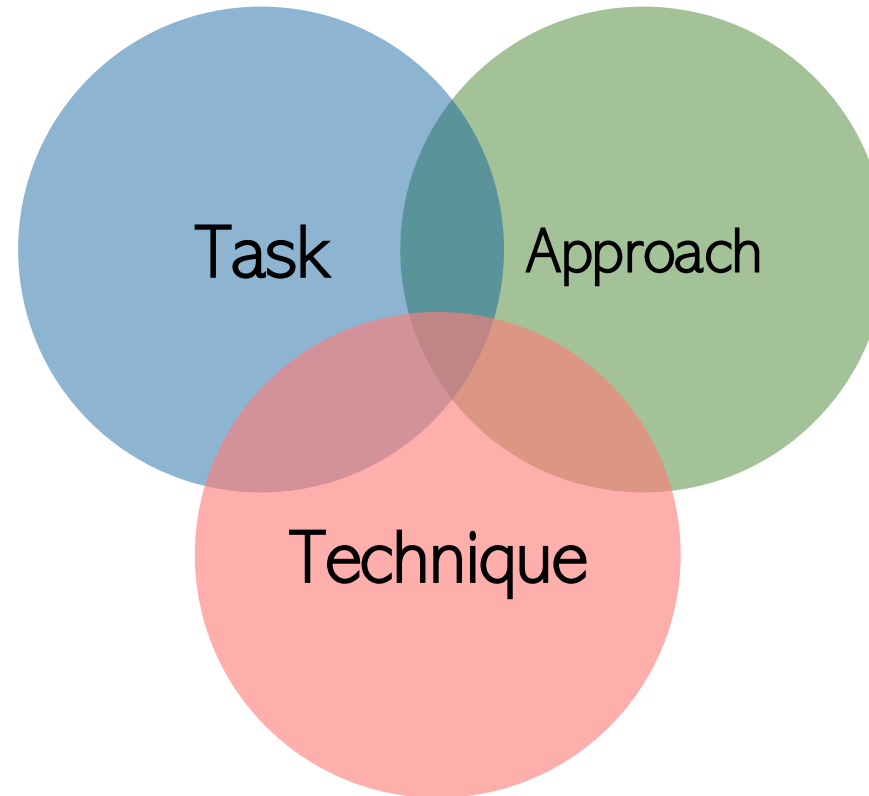
Name	Preferred Email	Bio
B02, Tue, 10:30AM		
Chowdhury, Reaz	reaz2@ualberta.ca	
Ebofin, Emmanuela	emmanuelaebofin@uvic.ca	
Kim, Alicia	20kimkayeon@gmail.com	I am a first-year Master's student in Computer Science, and I completed my undergraduate studies in Korea, majoring in AI and IoAs an undergraduate researcher, I explored reinforcement learning f robot navigation, focusing on enabling a robot to navigate toward a goal. I also explored system level integration by extending the conce of Software Defined Vehicles to robotics.
Motture, Kimberlee	kimberleegk@gmail.com	I am a second year PhD student under Dr Tzanetakis' supervision, a my Master's was in Music and Computer Science. I am fascinated b how sound can be manipulated with technology, and am taking this course to learn additional tools for doing that. I have a BMus.
Ryazanova, Uliana	uliana.a.ryazanova@gmail.com	I am a first-year Master's student in CS, with a bachelors in CS and certificate in Earth sciences. My interests lie in applied methods to environmental data- particularly in both representing and analyzing

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Outline: W02B: Topics Start

Classification

Regression
Clustering
Ranking (Recommendation)
Vector Representation and
Dimensionality Reduction
Language Modeling



Full/Self/Semi-Supervised Learning

Unsupervised Learning
Reinforcement Learning

k -Nearest Neighbor, Naive Bayes, Logistic Regression
Artificial Neural Networks, Graph Neural Network
Support Vector Machines
Principal Component Analysis
K-Means

Outline: W03A: Topics Start

I) Task: Classification

II) Supervision: Fully Supervised

III) Technique: k -Nearest Neighbor

III) Evaluation

Aletheia or Alethia /æliˈθaɪ.ə/ is truth or disclosure in philosophy. Originating in Ancient Greek philosophy, used for the first time in by Parmenides in his poem On Nature. He contrasts it with doxa (opinion). In Greek mythology, Aletheia was personified as a Greek goddess of truth, sincerity and unconcealment.

"the daughter of Zeus, unforgetting Truth"



Supervision: The Way of Truth

Natural Language:

Learning to assign an observation to one of a predefined set of classes.

Who tells the assignment is true, correct, valid, ...?

Formal Language:

Given a set of world observations $\mathbf{O}$ and a finite set $\mathbf{C}$, the task is to learn a mapping function $f: \mathbf{O} \rightarrow \mathbf{C}$, such that $f(o) = c; o \in \mathbf{O}, c \in \mathbf{C}$.

Who tells $f(o) \leftrightarrow c$ is true, correct, valid, ...?

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Supervision: The Way of Truth

Natural Language:

Learning to assign an observation to one of a predefined set of classes.

Who tells the assignment is true, correct, valid, ...?

Aletheia, the goddess of truth

God, in the Abrahamic religions as the all-powerful, all-knowing, eternal deity, and single creator of the universe

Omniscient /äm'niSHənt/: omni=all, scientia= knowledge or to know

The Oracle of Delphi

Teacher

Supervisor

Human (Expert) Annotator

Majority Vote on Human (Expert) Annotators

LLM-as-a-Judge



The oracle sat upon her tripod.
John Collier's *The Priestess of Delphi*, 1891

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Fully Supervised Classification

Natural Language:

Learning to assign an observation to ~~one~~ its **true class/label** in a predefined set of classes.

Formal Language:

Given a set of world observations and their true classes $\mathbf{O}^* = \{(o, c) \mid o \in \mathbf{O}, c \in \mathbf{C}\}$ where $\mathbf{C}$ is a finite set, the task is to learn a mapping function $f: \mathbf{O} \rightarrow \mathbf{C}$, such that for $(o, c) \in \mathbf{O}^*$, $f(o : \mathbf{O}^*) = c$.

Terminology

$\mathbf{O}^* = \{(o, c) \mid o \in \mathbf{O}, c \in \mathbf{C}\}$, labeled observations, experiences, or data set

Important:

f has **full access** to the entire $\mathbf{O}^*$



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Fully Supervised Classification: Preaching to the Converted?

We already have the labels $\mathbf{O}^* = \{(o, c) \mid o \in \mathbf{O}, c \in \mathbf{C}\}$,

→ Why learning a function f to tell us an observation's label?

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Fully Supervised Classification

We already have the labels $\mathbf{O}^* = \{(o, c) \mid o \in \mathbf{O}, c \in \mathbf{C}\}$,

→ Why learning a function f to tell us an observation's label?

→ Preaching to the Converted

→ Preaching to the Choir /'kwɪər/

→ Summarize the $(o \leftrightarrow c)$ associations in $\mathbf{O}^*$

E.g., classification of numbers into being even or odd

$\mathbf{O}^* = \{(n, c) \mid o \in \mathbf{N}, c \in \{\text{Odd}, \text{Even}\}\} \rightarrow f(n) = n \% 2$

A single operation as opposed to an entire set of numbers

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Fully Supervised Classification

We already have the labels $\mathbf{O}^* = \{(o, c) \mid o \in \mathbf{O}, c \in \mathbf{C}\}$,

→ Why learning a function f to tell us an observation's label?

→ Preaching to the Converted

→ Preaching to the Choir /'kwɪər/

→ Summarize the $(o \leftrightarrow c)$ associations in $\mathbf{O}^*$

→ Denoising $(o \leftrightarrow c)$ in $\mathbf{O}^*$

E.g., an even number is incorrectly labeled as odd

$(o: 24, c: \text{Odd}) \in \mathbf{O}^*$, but $f(24 : \mathbf{O}^*) = \text{Even}$

How? f sees the entire $\mathbf{O}^*$ and may understand that (majority of) numbers end with 4 are even.

Fully Supervised Classification

We already have the labels $\mathbf{O}^* = \{(o, c) \mid o \in \mathbf{O}, c \in \mathbf{C}\}$,

→ Why learning a function f to tell us an observation's label?

→ Preaching to the Converted

→ Preaching to the Choir /'kwɪər/

→ Summarize the $(o \leftrightarrow c)$ associations in $\mathbf{O}^*$

→ Denoising $(o \leftrightarrow c)$ in $\mathbf{O}^*$

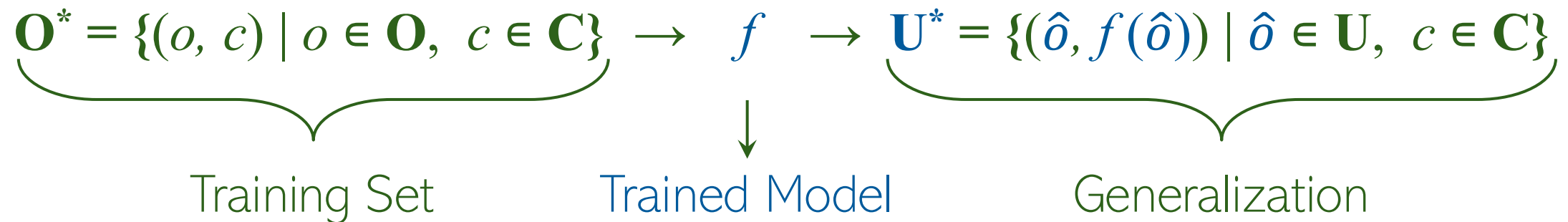
Our focus!

→ Inferring/predicting unseen/unknown/missing $(\hat{o} \leftrightarrow c) \notin \mathbf{O}^*$

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Fully Supervised Classification: $\mathbf{O}^* \rightarrow \mathbf{U}^*$

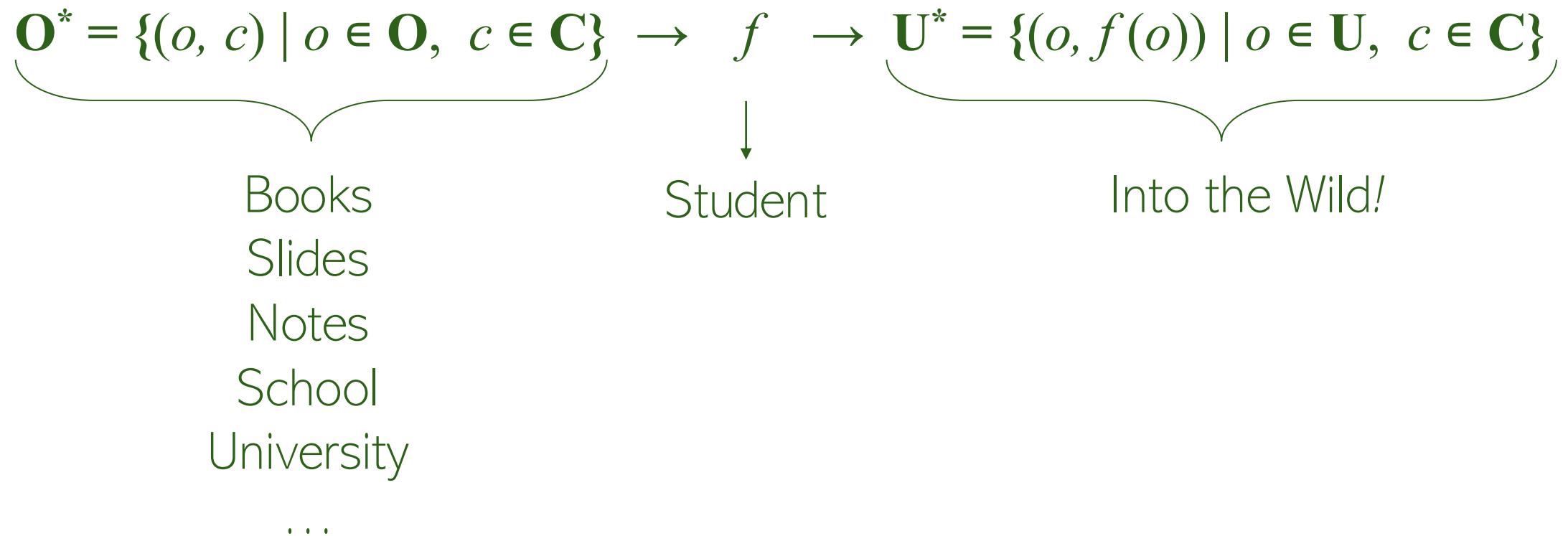
Generalize to all observations of the universe $\mathbf{U}$, where $\mathbf{O}^*$ is a tiny subset of it!



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Fully Supervised Classification: $\mathbf{O}^* \rightarrow \mathbf{U}^*$

Generalize to all observations of the universe $\mathbf{U}$, where $\mathbf{O}^*$ is a tiny subset of it!





Into the Wild, 2007, Sean Penn, Jon Krakauer
<https://www.youtube.com/watch?v=XZG1FzyB8DI>